

Experiment No- 7

Aim:

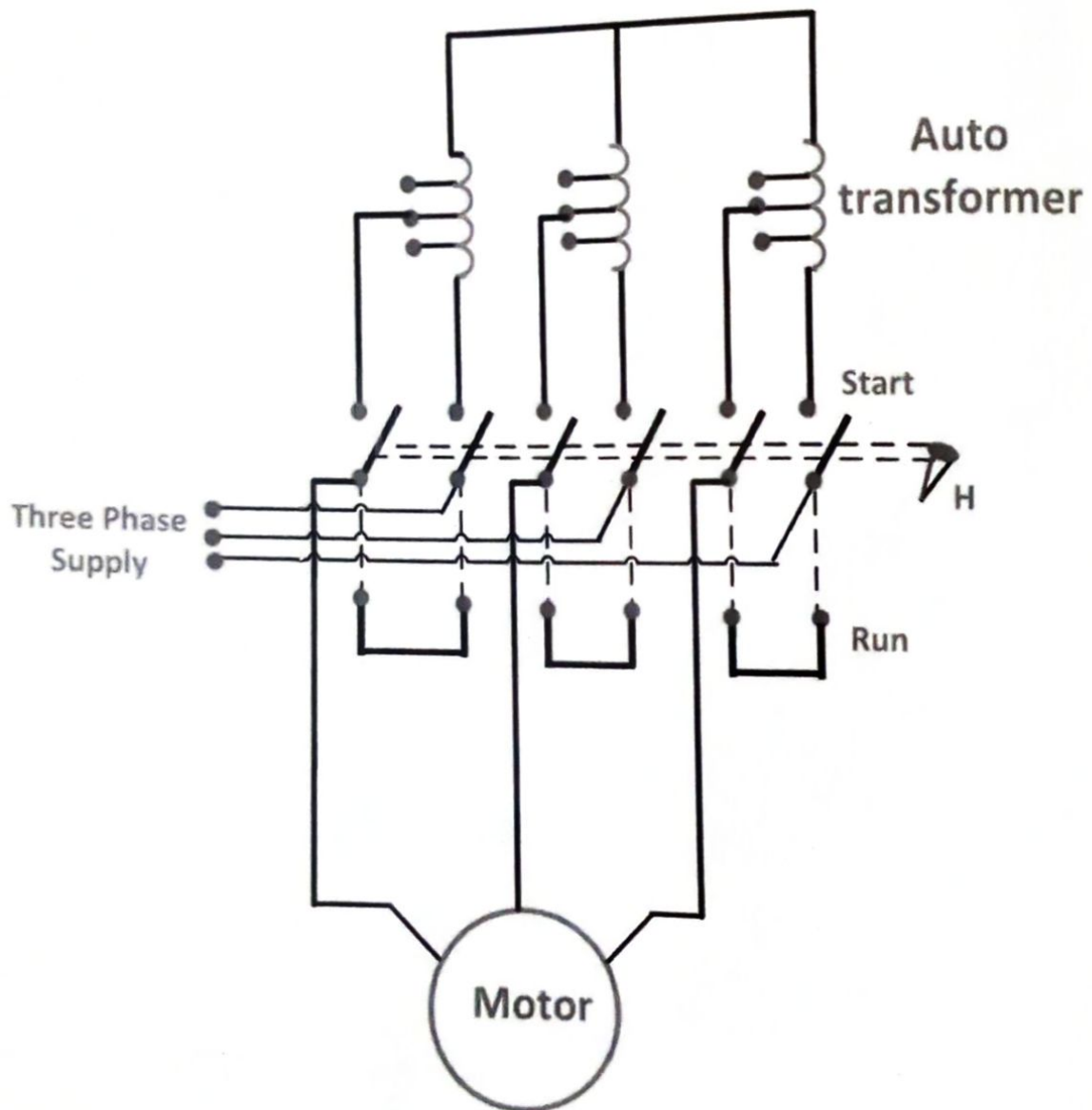
Study of various starters for 3- ϕ Induction motor.

Theory:

1. A starter is an electrical device which controls the electrical power for starting a motor.
2. A starter is necessary to limit this starting current to avoid damage to the motor as well as to other adjoining equipment.
3. Functions of Starters:
 - a) Current Limiting: - can be done by reducing the applied voltage or increasing the resistance in the circuit.
 - b) Overload Protection: - it is necessary because the motor draws more current during overload condition and it causes excessive heat in the windings. This extra heat reduces the motor's life and may cause burning of windings and hence fire.
 - c) Overheating Protection: - limit high current during overload.
4. Types of Starters:
 - a) Auto Transformer starter
 - b) Direct On-line starter
 - c) Star-Delta starter
 - d) Stator resistance starter

A) Auto-transformer starter:

Circuit diagram for Auto Transformer starter is given below:-



Circuit Globe

1. With an auto transformer starter, the transformer supplies a certain percentage of the primary voltage to the secondary of the transformer.
2. The auto transformer is connected in a star configuration.
3. The three tapped secondary coils of the transformer are connected to the three motor phases in this type of starter. This helps in reducing the voltage that is being applied to the motor terminals.

The Advantages of Using Auto Transformer Motor Starters:

- They can be used for manual speed control, but with limited options.

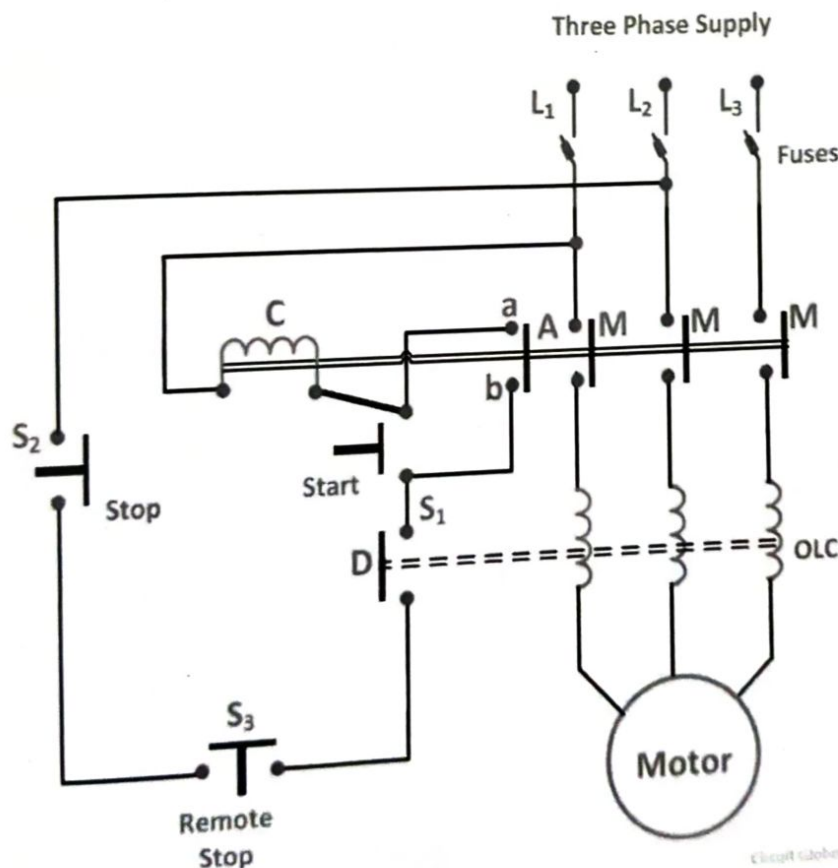
- They have extremely flexible starting characteristics.
- They have a high output torque.
- They are used for induction motors rated above 20HP.

Disadvantages of using Auto Transformer Motor Starters:

- The circuit is complex and involves a relatively expensive autotransformer
- Due to the size of the autotransformer starter, a much larger control, the panel is required which increases the price.
- If space is limited, it might not be possible to connect the starter to an existing machine due to its physical size.
- One of its major drawbacks is the frequency constraint of its operation. It is always necessary to know the operation frequency in order to select an autotransformer that is appropriately rated.

B) Direct on-line starter:

Circuit diagram for Direct-On-Line starter is given below:-



1. The DOL starter switches the supply directly on to the contacts of the motor.
2. As the starting current of an induction motor can be 6-8 times the running current the DOL starter is typically only used for motors with a rating of less than 5kW.

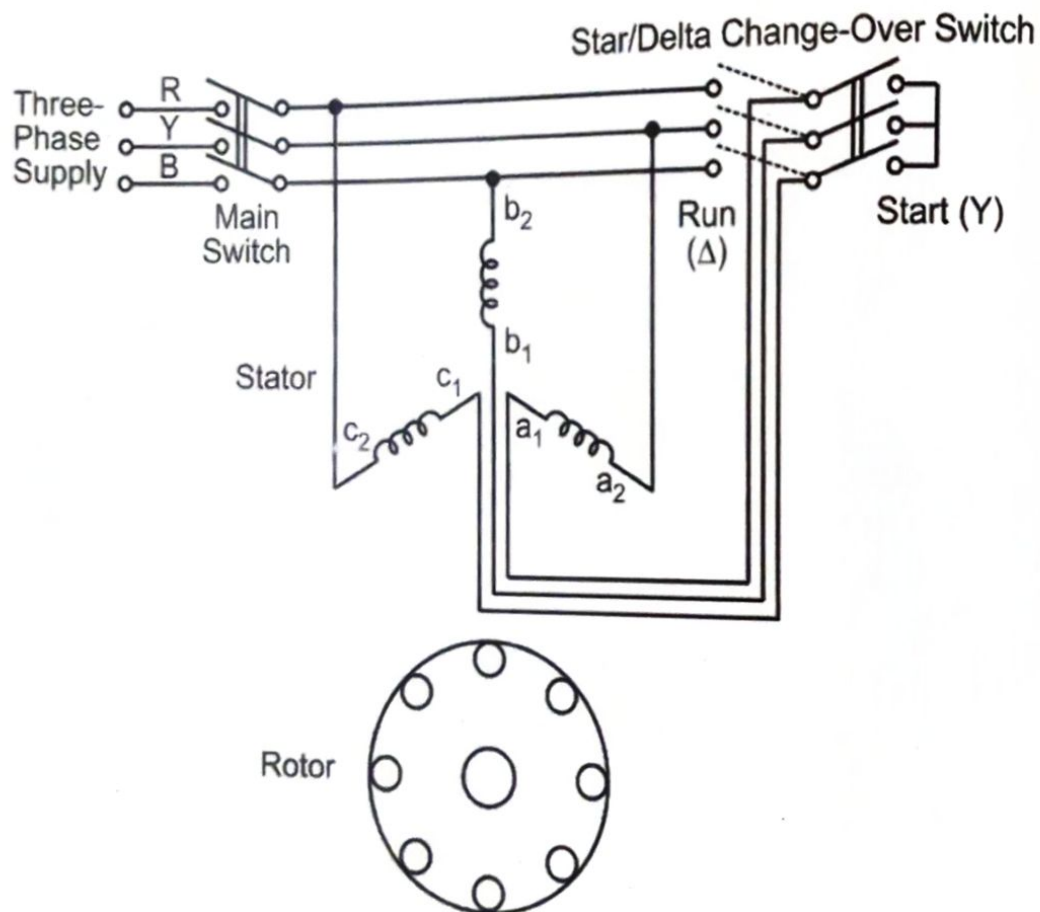
3. Large currents of short duration cause objectionable voltage drop in the power supply which cause undesirable dip in the supply line voltage; consequently, the operation of other equipment's is also affected.

The Advantages of Using Direct-On-Line Motor Starters:

- They have a compact design.
- They are cost-effective.
- They have a simple construction.
- They are used for motors rated upto 5HP.

C) Star – delta starter:

Circuit diagram for Star-Delta Starter is given below:-



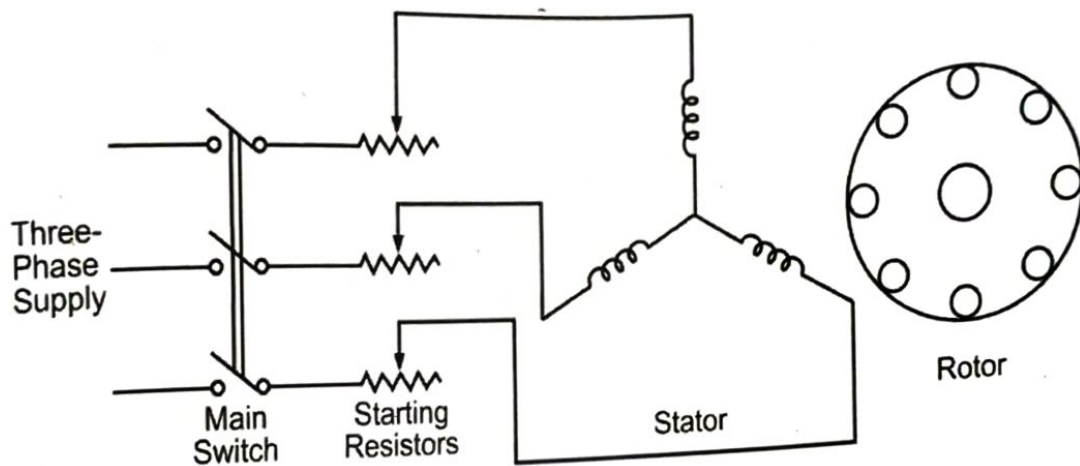
1. Compared to the other types of starters, the star delta starter is used on a large scale.
2. As the name suggests, the three windings are connected in a star connection in the star delta starters.
3. A certain time is set by the timer or any other controller circuit.
4. After this time is passed, the windings are then connected in the delta connection. The phase voltage in the star connection is reduced to 58%, and the total current drawn is 58% of normal current. This results in a reduced torque.

The Advantages of Using Star Delta Motor Starters:

- They are ideal for long acceleration times.
- They have a lower input surge current when compared to other starters.
- They have a simpler construction as compared to other starters.
- They are used for motors rated above 10HP.

D) Stator Resistance Starter:

Circuit diagram for Stator Resistance Starter is given below:-



1. The stator resistance starter consists of three resistors, which are connected in series with each phase of the stator windings.
2. At each resistor, a voltage drop is caused, so it becomes necessary to apply low voltage to each phase.
3. These resistances are set at the start or maximum position during the motor start stage.
4. The starting current is kept at a minimum in this type of starter. Also, the starting torque to the motor needs to be maintained.

The Advantages of Using Stator Resistance Motor Starters:

- They are suitable for use in speed control applications.
- They have extremely flexible starting characteristics.
- They provide smooth acceleration.